

Knowledge-Enhanced Domain Adaptation of Large Language Models for Bilingual Railway Vocational Education under Resource Constraints

Abstract

Purpose: The international expansion of railway technology has increased demand for bilingual vocational education, yet railway regulations, technical terminology and teaching materials remain difficult to access consistently across Chinese and English. Because most vocational institutions and local research teams work with limited computing resources, this study develops and evaluates a railway-domain large language model that answers bilingual railway questions on a single workstation.

Design/methodology/approach: The study addresses the problem with two methods. The first is a hybrid evidence-fusion method: BM25 lexical retrieval and BGE-M3 dense retrieval are fused by reciprocal rank fusion, and the top-ranked railway evidence passages are inserted into the prompt as the knowledge context for answering. The second is a vertical-domain adaptation method: completion-only QLoRA specialises Qwen2.5-7B-Instruct and GLM-4-9B-Chat-HF on reviewed railway question-answer pairs, with the base weights frozen in 4-bit NF4. Both methods were evaluated on a held-out bilingual railway set against the original models and an unadapted Qwen3-14B reference under an identical single-GPU boundary.

Findings: Evidence fusion produced the highest retrieval quality in both languages, reaching Evidence Recall@5 of 0.715 in Chinese and 0.708 in English, and fusing both language fields gave the best cross-language balance. Vertical fine-tuning raised held-out character-level F1 from 0.189 to 0.398 in Chinese and from 0.437 to 0.648 in English for Qwen2.5-7B. Combining both methods gave the best answer quality: Qwen QLoRA reached an answer F1 of 0.660 in Chinese and 0.651 in English under fused evidence, and adding that evidence improved answer F1 for every evaluated generator in both languages.

Originality/value: The study shows that lexical-dense evidence fusion and parameter-efficient vertical adaptation are complementary rather than substitutable mechanisms for building a domain model, especially in specialised vertical domains. Reporting the two methods under one resource boundary and one held-out protocol separates the gain obtained from retrieved railway knowledge from the gain obtained from domain-specific fine-tuning, and it gives a reproducible procedure for developing vertical-domain language models without multi-GPU training.

Keywords: railway domain large language model; bilingual vocational education; hybrid retrieval; reciprocal rank fusion; retrieval-augmented generation; QLoRA

1 Introduction

The international expansion of high-speed railway construction, operation and technical cooperation has increased demand for bilingual vocational education (Lawrence et al., 2019; Xu, 2018). Railway personnel and learners working across national and linguistic settings must understand operating rules, safety procedures, equipment terminology and teaching materials that are frequently written first in Chinese. The practical difficulty is therefore not limited to translating isolated terms. Educational support must connect a question posed in either language to the correct railway knowledge and then produce an answer that stays inside the boundaries of that knowledge.

General-purpose large language models can organise such knowledge fluently, but they are not trained on the terminology, evidence boundaries and procedural conventions of a specific railway corpus. Railway-oriented studies have built specialised models for safety, track management, prognostics, driver advice and interactive training (Zheng et al., 2023; Li et al., 2024; Song et al., 2025; Wang and Li, 2024; Luo et al., 2025; Li et al., 2025; Huang et al., 2026), and broader reviews continue to identify reliability and deployment as open problems for transport applications (Wandelt et al., 2024). Two weaknesses recur across that literature. First, the models are usually developed for one task and one language, so it is unclear whether the same approach transfers to a bilingual vocational setting where the same knowledge item must be answered in Chinese and in English. Second, the contribution of retrieved external knowledge is rarely separated from the contribution of model adaptation, which makes it difficult to tell which of the two mechanisms actually produced an observed improvement.

Computing availability is a second and equally binding constraint. Many vocational institutions and local research teams cannot maintain multi-GPU training clusters or rely continuously on commercial cloud inference. Full-parameter adaptation of large models is therefore difficult to reproduce and expensive to sustain in exactly the institutions that need a railway-domain assistant most. Quantised low-rank adaptation offers a realistic alternative: a 7B or 9B model can be specialised on a single 24 GB GPU while the base weights remain frozen. The resource constraint is consequently part of the research problem, not a secondary deployment detail.

This study addresses the problem with two methods.

The first method targets knowledge access and is referred to below as *evidence fusion*. BM25 lexical ranking and BGE-M3 dense ranking are computed in parallel and fused by reciprocal rank fusion, and the highest-ranked railway evidence passages are inserted into the prompt before generation. This method is what allows a question in either language to reach the correct reviewed railway record and to be answered from that record rather than from parametric memory.

The second method targets domain behaviour and is referred to below as *vertical adaptation*. Completion-only QLoRA trains low-rank adapters on reviewed railway question-answer pairs while the 4-bit quantised base model stays frozen, so the model learns railway answer vocabulary and answer structure without full-parameter training.

The two methods are then evaluated. Retrieval is scored by evidence-level metrics; adaptation is scored as a before-and-after comparison on held-out railway questions; the combined system is scored across five generators under identical cached evidence; and the resource cost of every condition is measured on the same workstation. This design answers a question that the existing railway-domain literature leaves open: whether improved knowledge access and improved domain behaviour are substitutes or complements, and what each costs inside a fixed hardware budget.

The study makes three contributions:

- It applies hybrid lexical-dense evidence fusion to bilingual railway vocational question answering and shows that fused retrieval gives the best evidence recall in both languages and that a bilingual field composition gives the best cross-language balance.
- It applies completion-only QLoRA vertical adaptation to two model families and quantifies the held-out bilingual gain with paired significance testing.
- It reports a controlled joint evaluation on the same held-out records showing that the two mechanisms are complementary but partially overlapping, and that vertical adaptation contributes the larger share of the gain on the same base model.

2 Related work

This section reviews the three lines of work that the present study builds on, states what each line actually contributed, and identifies the gap that remains.

2.1 Railway and transportation domain language models

A first line of work adapts general language models to railway and transportation tasks. Zheng *et al.* (2023) constructed a transportation-safety instruction corpus and fine-tuned a pre-trained model into a domain expert, demonstrating that domain instruction tuning changes task behaviour rather than only surface style. Li *et al.* (2024) built RSChat, an intelligent question-answering model over railway safety knowledge, which is the closest published antecedent to a railway vocational question-answering system. Song *et al.* (2025) specialised a language model for railway track management and reported domain-specific gains over a general model on track-management questions. Luo *et al.* (2025) developed a driver advisory system for high-speed trains in which the language model produces operational advice under timetable and signalling constraints. Li *et al.* (2025) evaluated a human-computer interactive training assistant for rail systems on dispatching and professional-training scenarios, which is the one prior study in this group that explicitly addresses railway education.

Two further strands extend the same idea. Huang *et al.* (2026) generated emergency operation schemes for urban rail transit train door systems with a retrieval-augmented model, showing that external evidence is needed when the required procedure is not reliably memorised. Wang and Li (2024) surveyed the potential applications, limitations and required solutions for language models in railway prognostics and health management, and Lahnalampi (2024) examined their use in rail design projects. At a broader level, Wandelt *et al.* (2024) reviewed language models for intelligent transportation and concluded that reliability, domain grounding and deployment cost remain unresolved.

What this line establishes is that railway tasks benefit from domain-specific language modelling, and that retrieval becomes necessary when the answer depends on a specific regulation or procedure. What it does not establish is a bilingual setting: the reviewed systems are built and evaluated in a single language, so the cross-language retrieval question, that is, whether the same evidence can serve a Chinese question and an English question equally well, is left open.

2.2 Retrieval-augmented generation and hybrid evidence fusion

A second line of work supplies the knowledge-access mechanism. Lewis *et al.* (2020) introduced retrieval-augmented generation, in which a parametric generator is conditioned on documents retrieved from an external non-parametric index. This organisation is directly relevant to railway vocational education because regulations, terminology records and teaching materials are revised over time, and an external index can be updated without retraining the generator.

Within that framework, the retrieval component has been developed along two complementary directions. Robertson and Zaragoza (2009) formalised the probabilistic relevance framework and its BM25 instantiation, which remains strong when a query and a document share exact terms such as equipment names, regulation identifiers and standard codes. Chen *et al.* (2024) introduced BGE-M3, a multilingual embedding model that produces dense representations for semantically related but lexically different text, which matters when a Chinese and an English question express the same need with different wording. Because the two retrievers score on incommensurable scales, Cormack *et al.* (2009) proposed reciprocal rank fusion, which merges ranked lists using only rank positions and therefore requires no score normalisation. Hübscher *et al.* (2021) addressed how knowledge and process descriptions should be represented and presented in dynamic, communication-intensive environments, which is relevant to how a domain record should be structured so that it stays usable by both a human reviewer and a downstream model. In the legal-document setting, Yang *et al.* (2024) surveyed question answering over long professional documents and showed that retrieval strategy and answer verification are the two decisive design choices in that genre. Chang *et al.* (2024) surveyed language-model evaluation and argued that distinct capabilities require distinct metrics instead of a single aggregate score.

This line therefore supplies the first method used here: lexical and dense retrieval fused at rank level. Its evaluations, however, are predominantly monolingual and mostly assume a single-language index. The question of how the index should be composed when every knowledge item exists in two languages, and whether a bilingual field composition beats a language-specific one, is not settled by this literature.

2.3 Parameter-efficient vertical-domain adaptation

A third line of work supplies the adaptation mechanism. Dettmers *et al.* (2023) introduced QLoRA, which back-propagates through a frozen 4-bit NF4 quantised base model into low-rank adapters, reducing training memory enough to fine-tune multi-billion-parameter models on a single device. Ding *et al.* (2023) and Wang *et al.* (2025) surveyed parameter-efficient fine-tuning and catalogued the trade-offs between adapter capacity, trainable parameter count and task performance. Lv *et al.* (2024) examined full-parameter fine-tuning under limited resources and documented how sensitive such training is to quantisation, sequence length and optimiser state, which is why the present study fixes those settings explicitly rather than leaving them to defaults.

A consistent warning runs through this literature: adaptation is not automatically beneficial. Tian *et al.* (2024) showed that fine-tuning can improve factuality, but Barnett *et al.* (2024) demonstrated that fine-tuning frequently underperforms simpler alternatives when the evaluation protocol is not controlled. In the educational setting that motivates this study, Kasneci *et al.* (2023) and UNESCO (2023) identified factual reliability, learner over-reliance and opaque provenance as the main risks of generative models in teaching. Schwartz *et al.* (2020) argued more generally that predictive quality should be reported together with its computational cost, and that reporting quality alone misrepresents the value of an efficient method. Adapter size by itself therefore does not establish low inference cost, and a before-and-after design plus measured resource use is required before an adaptation gain can be claimed.

2.4 Research gap

The three lines above supply a domain motivation, a knowledge-access mechanism and an adaptation mechanism, but they have not been combined under one controlled protocol. Railway-domain studies are essentially monolingual and usually report a single mechanism. Hybrid-retrieval studies are essentially monolingual in their index design and rarely measure resource cost. Parameter-efficient adaptation studies rarely test whether the observed gain survives once external evidence is also supplied.

The present study fills that gap by applying both methods to the same bilingual railway corpus and evaluating them on the same held-out records under one hardware boundary. This allows three questions to be answered together: whether fused bilingual retrieval improves evidence access over either retriever alone, whether completion-only QLoRA improves held-out railway question answering, and whether the two gains are additive or overlapping when the adapted models also receive fused evidence.

3 Method

3.1 Problem definition and resource boundary

The task is bilingual railway question answering over reviewed regulations, terminology records and vocational teaching material. Let $\mathcal{D}$ denote the governed knowledge base of railway records and let q be a question written in Chinese or English. The system must return an answer $\hat{y}$ in the language of q :

$$\hat{y} = f_{\theta_0, \theta_A}(q, \mathcal{R}(q, \mathcal{D})), \quad (1)$$

where $\mathcal{R}(q, \mathcal{D})$ is the evidence returned by the retrieval function and f is a generator whose base weights θ_0 are frozen and whose adapter weights θ_A carry the domain adaptation. Equation (1) states the two methods of this study directly: $\mathcal{R}$ is the evidence-fusion method, and θ_A is the vertical-adaptation method. When $\mathcal{R}(q, \mathcal{D}) = \emptyset$ the generator answers from parametric memory alone, which is the no-retrieval condition used as a baseline throughout.

The resource boundary is one NVIDIA RTX 3090 GPU with 24 GB of memory and 32 GB of system RAM, with no multi-GPU training and no required cloud inference. Writing M for the peak GPU memory of a condition, every configuration must satisfy

$$M \leq 24 \text{ GB}, \quad (2)$$

and the reported scores are only comparable because Equation (2) holds for all of them. This constraint is what makes NF4 quantisation and low-rank adaptation necessary rather than optional, since full-parameter training of the selected 7B and 9B models exceeds the boundary.

3.2 Overall framework

Figure 1 shows how the two methods are implemented and where they meet. The framework reads from left to right in four stages. Stage one prepares the governed bilingual inputs: each knowledge record carries a Chinese and an English field that share one knowledge-pair identifier, and an admissibility filter restricts retrieval to approved records while the held-out test records are excluded from the index. Stage two encodes a query and a candidate passage with the same frozen BGE-M3 encoder, which is what makes a Chinese query comparable with an English passage. Stage three runs the two retrievers in parallel and fuses their rankings, which is the evidence-fusion method. Stage four conditions one shared generator on the serialised context, where the QLoRA adapter is attached to the frozen quantised base model, which is the vertical-adaptation method.

The figure also shows the property that the experiments are designed to test: the two methods act on different parts of the same system. Evidence fusion changes what the generator is allowed to see, whereas vertical adaptation changes how the generator answers. Because the two mechanisms are separate in the architecture, their contributions can be measured separately by holding one of them fixed and varying the other.

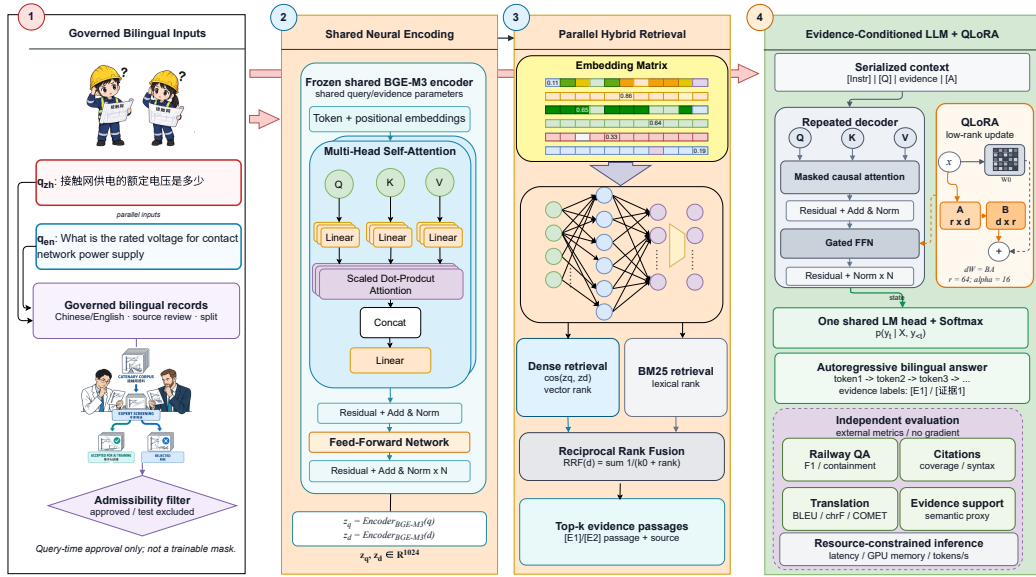


Figure 1: Bilingual railway question answering with two methods. Left to right: (1) governed bilingual inputs, where a Chinese query q_{zh} and an English query q_{en} address the same knowledge item and an admissibility filter keeps only approved pairs; (2) shared neural encoding, where one frozen BGE-M3 encoder maps the query and the candidate evidence into the same 1024-dimensional space through token and positional embeddings, multi-head self-attention and a feed-forward network; (3) parallel hybrid retrieval with evidence fusion, where a dense retriever scores cosine similarity and a BM25 retriever scores lexical overlap, and reciprocal rank fusion merges the two rankings into a top- k evidence set; and (4) evidence-conditioned generation with vertical adaptation, where the serialised context $[\text{Instr}] | [Q] | \text{evidence} | [A]$ is decoded by a repeated causal decoder whose frozen weights W_0 receive a low-rank update $dW = BA$ (rank 64, alpha 16), followed by an independent evaluation block that sits outside the model and carries no gradient. That block reports the railway question-answering scores used as the primary results of this study, and it also marks the downstream uses that the system is designed to support once deployed, namely citing a generated answer against its evidence, translating an answer between Chinese and English, checking whether its claims are supported by the retrieved passages, and measuring latency and GPU memory under the resource-constrained inference boundary. The upper path is the evidence-fusion method and the lower path is the vertical-adaptation method; both meet at the same generator so that their contributions can be distinguished. Source: Authors’ own work.

3.3 Method one: bilingual hybrid retrieval with evidence fusion

The first method solves the knowledge-access problem. Railway questions mix exact identifiers with paraphrased technical descriptions, and no single retriever handles both. The method therefore computes two rankings in parallel and merges them at rank level.

The lexical ranking scores a candidate passage d against query q with BM25 (Robertson and Zaragoza, 2009):

$$\text{BM25}(q, d) = \sum_{t \in q} \text{IDF}(t) \cdot \frac{f(t, d)(k_1 + 1)}{f(t, d) + k_1 \left(1 - b + b \frac{|d|}{\text{avgdl}}\right)} \cdot (1 + 0.05 \min(f(t, q), 3)), \quad (3)$$

where $f(t, d)$ is the frequency of term t in d , $f(t, q)$ is its frequency in the query, $|d|$ is the passage length, avgdl is the average passage length in the index, and k_1 and b control term-frequency saturation and length normalisation. The implementation sets $k_1 = 1.5$ and $b = 0.75$, and the final factor mildly up-weights a query term that occurs more than once, saturating at three occurrences. The corpus is small enough that the index is held in memory, so no external search library is involved. Then

$$\text{IDF}(t) = \ln \left(\frac{N - n(t) + 0.5}{n(t) + 0.5} + 1 \right) \quad (4)$$

with N the number of indexed passages and $n(t)$ the number containing t . Equation (3) rewards exact overlap, which is what preserves equipment names, clause numbers and standard codes that a dense encoder may blur.

The dense ranking embeds the query and the passage with BGE-M3 (Chen et al., 2024) and scores them by cosine similarity:

$$s_{\text{dense}}(q, d) = \frac{\mathbf{e}_q^\top \mathbf{e}_d}{\|\mathbf{e}_q\|_2 \|\mathbf{e}_d\|_2}, \quad (5)$$

where $\mathbf{e}_q, \mathbf{e}_d \in \mathbb{R}^{1024}$. Equation (5) rewards semantic equivalence, which is what recovers an English passage for a Chinese query that shares no surface term with it.

Because Equations (3) and (5) are not on a common scale, the two rankings are merged by reciprocal rank fusion (Cormack et al., 2009):

$$\text{RRF}(d) = \sum_{r \in \{\text{BM25}, \text{dense}\}} \frac{1}{k_0 + \text{rank}_r(d)}, \quad (6)$$

where $\text{rank}_r(d)$ is the position of d in ranking r and k_0 is a smoothing constant. A passage absent from a retriever’s candidate list contributes zero for that retriever. Equation (6) depends only on rank position, so no score calibration between the lexical and dense retrievers is needed. The candidate pool is fixed at 50 per retriever and $k_0 = 60$, and the fusion is applied before the final cut-off is chosen.

The fused list is truncated to the top- k passages, and these are serialised into the prompt as the knowledge context:

$$\mathcal{R}(q, \mathcal{D}) = \text{top-}k \text{ RRF}(d), \quad d \in \mathcal{D}_{\text{adm}} \quad (7)$$

where $\mathcal{D}_{\text{adm}}$ is the admissible subset of the knowledge base, and the prompt is assembled as

$$x = [\text{Instr}] \mid [q] \mid [E_1] \dots [E_k] \mid [A], \quad (8)$$

where each E_i is rendered into the prompt as its numbered label, its source document and the passage text, while the retrieval object also carries the record identifier, source document and task type that the evaluation needs. Equation (8) is the point at which the first method hands control to the generator: the tags $[E_i]$ give the model an addressable evidence set, and the experiment varies k from one to eight to measure how much evidence is worth supplying.

Two design choices distinguish this implementation. First, the same encoder is used for the query and for the evidence, so no cross-lingual alignment step is required at query time. Second, held-out records are removed before ranking: the lexical index is built with test-split and reserved RAG-test records excluded, and the dense query carries the same exclusion in its database predicate. This guarantees that a held-out record cannot be returned as evidence for its own question. The approved-only condition is a separate control that is enforced while candidates are returned rather than assumed of the corpus, and the two controls are reported separately in the results.

3.4 Method two: completion-only QLoRA vertical adaptation

The second method solves the domain-behaviour problem. A general model produces fluent answers that use the wrong register, omit the procedural detail expected in a railway answer, or answer a regulation question by paraphrase instead of by clause content. The method corrects this by training low-rank adapters on reviewed railway question-answer pairs while the base model stays frozen.

For a pre-trained weight matrix $W_0 \in \mathbb{R}^{d \times k}$, the adapted forward pass is

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} B A x, \quad (9)$$

where $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$ are the trainable low-rank factors, r is the rank and α is a fixed scaling constant (Dettmers et al., 2023). Only A and B receive gradients, so the trainable parameter count is $r(d+k)$ rather than dk ; with $r = 64$ this is a small fraction of the full matrix, which is what makes the method fit inside Equation (2).

The frozen base weights are stored in 4-bit NF4 format. A block of weights is dequantised with per-block scale and zero-point constants before the forward pass:

$$W_0 \approx s_b (q_b - c_b), \quad (10)$$

where q_b is the quantised block, and s_b and c_b are the block scale and offset. Equation (10) is a lossy representation of the base weights, and the adapter in Equation (9) is what restores task capacity at that reduced precision.

Supervision is applied to answer tokens only. With $m_t = 1$ for an answer token and $m_t = 0$ for a prompt token, the training objective is

$$\mathcal{L}(\theta_A) = -\frac{1}{\sum_t m_t} \sum_t m_t \log p_{\theta_0, \theta_A}(y_t \mid y_{<t}, x), \quad (11)$$

where θ_0 denotes the frozen NF4 base weights and $\theta_A = \{\theta_A^{(l)}\}_{l=1}^L$ the trainable adapters in their respective layers. Equation (11) is the reason the method is called completion-only: masking the prompt positions prevents the model from spending capacity on reproducing instructions and concentrates it on the railway answer, and it keeps the training signal identical to the inference-time task defined in Equation (1).

The adaptation corpus is built only from approved records and is split by knowledge-pair identifier, task type and source document. Chinese and English variants of one knowledge item always receive the same partition assignment, so a bilingual pair cannot straddle the training and held-out boundary, and no knowledge-pair identifier is shared across partitions. Completion-only masking labels every prompt token with -100 ; the loss is computed only on answer tokens.

3.5 Evaluation protocol

Retrieval is scored at the evidence level rather than the record level, because a different admissible record may legitimately contain the evidence required to answer the question. Evidence-equivalent Recall@ k is one when at least one of the first k candidates satisfies the matching rule:

$$\text{Recall@}k = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbf{1}[\exists d \in \text{top-}k(q) : \text{match}(d, q)], \quad (12)$$

where $\text{match}(d, q)$ holds when the candidate shares the record identifier with the gold evidence, when the whitespace-normalised gold and candidate evidence contain one another in either direction, or when the candidate comes from the same source document and contains the gold answer. Mean reciprocal rank is computed over the rank of the first matching candidate:

$$\text{MRR} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{\text{rank}_q^*}, \quad (13)$$

where rank_q^* is the rank of the first matching candidate for q , and contributes zero when no candidate matches.

Generation is scored by character-level overlap between the prediction $\hat{y}$ and the reference y , computed on multisets of characters after answer normalisation and whitespace removal:

$$P = \frac{|\hat{y} \cap y|}{|\hat{y}|}, \quad R = \frac{|\hat{y} \cap y|}{|y|}, \quad F1 = \frac{2PR}{P + R}. \quad (14)$$

In the retrieval-augmented experiments the same overlap is computed over individual Chinese characters and lower-cased Latin and alphanumeric tokens. The two definitions are reported within their own settings and are not treated as interchangeable absolute scores. Reference containment is reported alongside $F1$ as a stricter indicator of whether the normalised reference occurs in the prediction.

All generator comparisons use paired samples. Differences between conditions are reported with two-sided paired Wilcoxon tests, the paired effect size

$$d_z = \frac{\bar{d}}{s_d}, \quad (15)$$

where $\bar{d}$ is the mean of the per-sample differences and s_d their standard deviation, and Holm correction across the pre-specified comparison family. Mean metrics are accompanied by 2,000-resample bootstrap 95 per cent confidence intervals.

Resource use is measured over 30 Chinese short questions, 30 English short questions and 30 regulation-oriented long-context questions, with one warm-up prompt at the start of each condition and three measured repetitions, giving 270 measurements per condition. Reported quantities are peak and static reserved GPU memory, first-token latency, mean end-to-end latency and generation throughput.

3.6 Experimental configuration

Table I defines the single-workstation boundary and the shared settings. All retrieval, adaptation and generation conditions were executed inside this boundary without multi-GPU training or cloud inference.

The two adapted models require different adapter placement because their attention and feed-forward modules are not named alike. Qwen2.5-7B exposes separate gate and up projections, whereas GLM-4-9B

Table I: Single-workstation implementation and experimental configuration.

Component	Formal setting
Database	PostgreSQL 16.14; pgvector 0.8.4
Embedding	BAAI/bge-m3; 1,024 dimensions; frozen shared encoder
Adapted generators	Qwen2.5-7B-Instruct; GLM-4-9B-Chat-HF
Reference generator	Qwen3-14B through Ollama 0.19.0 (unadapted)
Retrieval	BM25 and dense, 50 candidates each; RRF with $k_0 = 60$
Training	NF4 QLoRA; rank 64; alpha 16; dropout 0.05; one epoch; seed 42
Optimisation	Learning rate $2e-4$; batch size 1; 16 gradient accumulation steps; warm-up ratio 0.03
Sequence length	1,024 tokens maximum
Decoding	Greedy; temperature 0; top- p 1
Hardware boundary	One RTX 3090 24 GB; 32 GB RAM; no multi-GPU or required cloud inference
Statistical analysis	2,000 bootstrap resamples; paired Wilcoxon; Cohen’s d_z ; Holm correction
Software	Linux; Python 3.11.15; PyTorch 2.11.0; Transformers 5.12.1; PEFT 0.18.1; bitsandbytes 0.49.2; Ollama 0.19.0

Table II: Vertical adaptation configuration and trainable parameter counts.

Model	Checkpoint	Quantisation	Target modules	Trainable parameters	Share
Qwen2.5-7B	Qwen/Qwen2.5-7B-Instruct	NF4 4-bit	q, k, v, o, up, down, gate	161,480,704	3.58%
GLM-4-9B	THUDM/glm-4-9b-chat-hf	NF4 4-bit	q, k, v, o, down, gate_up	190,382,080	3.45%

uses a fused gate-up projection, so the target module list is model-specific. Table II records the resulting adaptation matrix and the trainable parameter counts.

The share column reports trainable parameters as a proportion of the parameters visible to the quantised training process, not as a proportion of the original full-precision model. The base model architectures and their post-training lineage are documented in the corresponding technical reports (Qwen Team, 2024; GLM Team, 2024), and the unadapted reference model in (Qwen Team, 2025).

4 Results

4.1 Corpus and experimental sample

The governed knowledge base contains 37,661 approved records and 37,109 complete bilingual records, in which a Chinese and an English field share one knowledge-pair identifier. The sources are railway terminology, operating and safety regulations, and vocational textbook or concept material. From the approved bilingual pairs, 7,614 were selected under a task cap and split 80/10/10 by knowledge-pair identifier into 12,178 training, 1,524 validation and 1,526 held-out records, balanced equally between the two languages. Pair overlap is zero across all three partitions. The held-out partition contains 763 knowledge pairs across four source documents; three quarters of the terminology pairs reach the per-task cap, so the frozen sample is dominated by terminology, regulation and textbook tasks in that proportion.

Two evaluation samples are drawn from the held-out partition. The standalone bilingual question-answering sample uses all 763 pairs in both languages. The cross-source retrieval-augmented sample uses 400 pairs and 800 language-specific queries, covering 150 regulation, 127 terminology and 123 textbook or concept pairs across 17 task types. The same 400 pairs and the same cached evidence are used for every generator, so generator differences cannot arise from differences in retrieved context.

4.2 Evidence fusion improves retrieval in both languages

Table III reports the retrieval comparison on the 400-pair cross-source sample. Three observations follow from it.

First, the two retrievers are not interchangeable and their weaknesses are language-dependent. BM25 is the strongest single retriever for English, reaching an Evidence Recall@1 of 0.570 and an MRR of 0.620, which reflects the many English queries that share exact terminology with their evidence. Dense retrieval is

Table III: Retrieval quality and latency on 400 held-out knowledge pairs.

Retrieval	Language	Evidence R@1	Evidence R@3	Evidence R@5	MRR	Mean latency (ms)
BM25	Chinese	0.520	0.595	0.620	0.560	69.5
Dense	Chinese	0.518	0.653	0.690	0.588	134.4
Fused	Chinese	0.570	0.675	0.715	0.628	217.6
BM25	English	0.570	0.670	0.685	0.620	59.0
Dense	English	0.463	0.553	0.580	0.509	134.4
Fused	English	0.570	0.668	0.708	0.620	209.2

the stronger single retriever for Chinese at recall levels above one, reaching 0.690 Evidence Recall@5 against 0.620 for BM25, which reflects Chinese queries that are paraphrases rather than term matches. Neither retriever dominates, which is the precondition for fusion to be useful.

Second, fusion improves on both. The fused ranking attains the highest Evidence Recall@5 in Chinese (0.715) and in English (0.708) and the highest MRR in both languages (0.628 and 0.620). The gain is largest where the single retrievers disagree most: in Chinese, fusion raises Recall@1 from 0.520 to 0.570 while the dense retriever alone was 0.518, meaning the lexical ranking still contributed candidates that the dense ranking had missed.

Third, the improvement is not free. Mean retrieval latency rises from 69.5 ms for BM25 and 134.4 ms for dense retrieval to 217.6 ms for the fused Chinese condition, so fusion is a quality-latency trade-off rather than a uniform efficiency gain. Filtering to approved records changes nothing in this frozen experiment, because the governed corpus holds 37,661 approved records out of 37,664, so the admissible index and the approved index are the same index in practice; the filter matters for correctness of the protocol rather than for the measured scores.

Figure 2 shows how the final evidence cut-off interacts with quality and latency. Recall rises monotonically with k , from 0.570 at $k = 1$ to 0.743 in Chinese and 0.723 in English at $k = 8$; the largest single step is between $k = 1$ and $k = 3$, after which the curve flattens. Retrieval latency is almost flat across the same range, staying near 218 ms in Chinese and 209 ms in English, because each setting reuses the same 50-candidate pool from each retriever and only the truncation point changes. The evidence supplied to the generator can therefore be increased at essentially no additional retrieval cost, although a longer context still consumes generation time and memory.

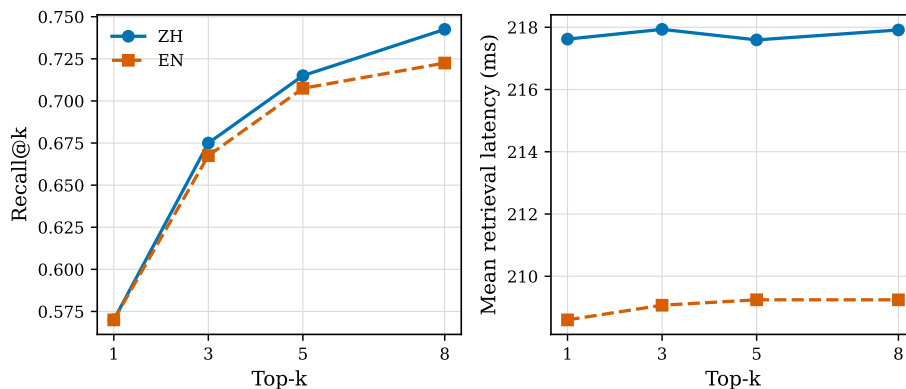


Figure 2: Effect of the final evidence cut-off on fused retrieval. Recall rises from 0.570 at top-1 to 0.743 in Chinese and 0.723 in English at top-8, whereas mean retrieval latency remains near 218 ms and 209 ms because the upstream candidate pool is fixed at 50 passages per retriever. Increasing the fused cut-off therefore improves evidence coverage without materially changing search time. Source: Authors’ own work.

4.3 Bilingual index composition

Because every knowledge item exists in both languages, the index can be built from the Chinese field, from the English field, from the source evidence field, or from both language fields. The four compositions were compared on the same sample, and the result argues for the bilingual index.

Under fused retrieval, the bilingual index reaches the highest balanced mean across languages, with an Evidence Recall@5 of 0.718 in Chinese and 0.675 in English, giving a mean of 0.696. The English-field index is better on English alone (0.713) but much weaker in Chinese (0.560), and the Chinese-field index is better in Chinese (0.653) but weaker in English (0.530). The source-evidence index is the weakest overall

(0.638 Chinese, 0.448 English). A language-specific index therefore wins on its own language and loses on the other, which is the wrong trade-off for a bilingual vocational assistant, where a learner may ask in either language about the same knowledge item. Indexing both language fields is what keeps the two languages within a comparable range without a per-language index switch.

4.4 Vertical adaptation improves held-out railway question answering

Table IV reports the before-and-after comparison on the full 763-pair held-out set, evaluated without retrieval so that the measurement isolates the second method.

Table IV: Held-out railway question answering before and after vertical adaptation (character-level F1, $n = 763$ per cell).

Model	Language	Original F1 (95% CI)	Adapted F1 (95% CI)	Absolute gain	Cohen's d_z
Qwen2.5-7B	Chinese	0.189 (0.175–0.204)	0.398 (0.376–0.422)	+0.209	0.569
Qwen2.5-7B	English	0.437 (0.419–0.455)	0.648 (0.632–0.664)	+0.211	0.634
GLM-4-9B	Chinese	0.192 (0.179–0.207)	0.410 (0.384–0.436)	+0.218	0.614
GLM-4-9B	English	0.498 (0.481–0.515)	0.552 (0.522–0.582)	+0.054	0.146

All four paired gains remain significant after Holm correction, with adjusted p -values below 0.01 in every condition. The Chinese gain is large for both families, roughly doubling the original score: Qwen2.5-7B rises from 0.189 to 0.398 and GLM-4-9B from 0.192 to 0.410. The English behaviour differs by model. Qwen2.5-7B improves substantially, from 0.437 to 0.648, whereas GLM-4-9B improves only from 0.498 to 0.552, with a small paired effect size of 0.146. Pooling the two languages would therefore hide a real difference between the models, and the two languages are reported separately throughout.

The relative pattern is informative in a second way. Before adaptation, GLM-4-9B is the stronger English model and the two models are equivalent in Chinese. After adaptation, Qwen2.5-7B is the strongest model in both languages, and Qwen’s bilingual gap narrows from 0.248 to 0.250 in absolute terms but from a 131 per cent to a 63 per cent relative advantage for English. Adaptation therefore helps most where the original model was weakest, which is consistent with the method correcting missing domain vocabulary rather than uniformly raising all outputs.

Figure 3 shows the optimisation behaviour that produced these gains. Both models reduce training loss steeply at the start of the single epoch and then flatten, and the end-of-epoch validation measurements sit close to the final training range, which indicates that the one-epoch schedule under the resource limit was sufficient to fit the adaptation set without a visible divergence between the curves. Qwen ends below GLM on both the training and the validation curve, which is consistent with its stronger held-out result.

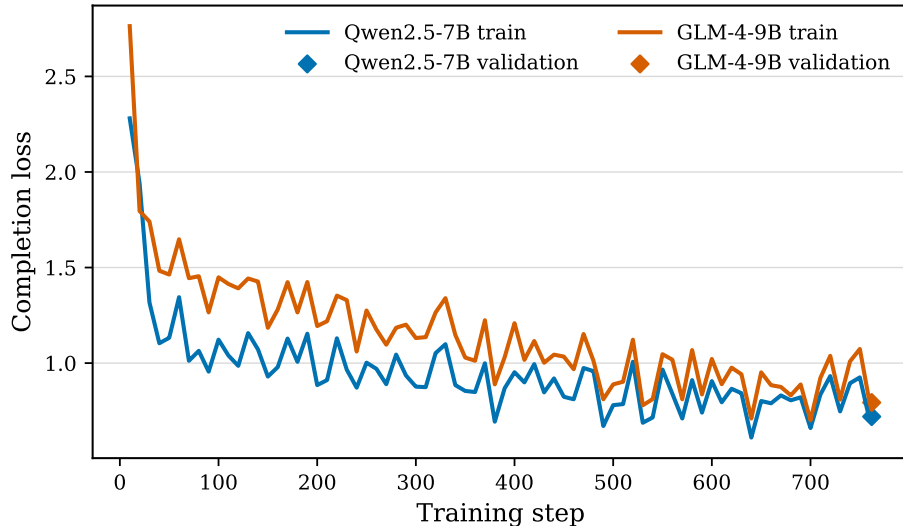


Figure 3: Completion-only QLoRA optimisation over one epoch. Training loss falls rapidly for both models and then flattens; diamonds mark the end-of-epoch validation measurement, which stays close to the final training range for both. Qwen2.5-7B finishes below GLM-4-9B on both curves, consistent with its stronger held-out result. The curves support successful optimisation inside the resource limit but do not establish behaviour under longer schedules or repeated seeds. Source: Authors’ own work.

4.5 Joint evaluation of evidence fusion and vertical adaptation

The two methods were then applied together. Five generators were evaluated under three retrieval conditions on the identical 400-pair sample with identical cached evidence: no retrieval, BM25 retrieval, and fused approved retrieval. Table V reports the comparison between no retrieval and fused evidence, and reports the adapted models alongside the original ones and the larger unadapted reference.

Table V: Answer quality under no retrieval and under fused evidence ($n = 400$ queries per cell).

Generator	No retrieval (ZH/EN)	BM25 (ZH/EN)	Fused (ZH/EN)	Fusion gain (ZH/EN)	Holm p (ZH/EN)
Qwen2.5-7B original	0.262 / 0.306	0.406 / 0.433	0.402 / 0.440	+0.141 / +0.133	1.9e-39 / 2.3e-42
GLM-4-9B original	0.230 / 0.269	0.291 / 0.314	0.298 / 0.318	+0.067 / +0.048	1.1e-22 / 9.6e-31
Qwen2.5-7B QLoRA	0.630 / 0.574	0.674 / 0.644	0.660 / 0.651	+0.030 / +0.078	1.3e-06 / 1.4e-11
GLM-4-9B QLoRA	0.398 / 0.331	0.482 / 0.355	0.476 / 0.351	+0.077 / +0.020	3.2e-09 / 9.4e-03
Qwen3-14B unadapted	0.295 / 0.336	0.439 / 0.464	0.442 / 0.454	+0.147 / +0.118	6.7e-32 / 7.6e-29

Adding fused evidence improves answer F1 over no retrieval for every generator and language, and all ten comparisons remain significant after Holm correction. The magnitude of the gain, however, depends strongly on the generator. The original Qwen2.5-7B gains 0.141 in Chinese and 0.133 in English, and the unadapted Qwen3-14B gains 0.147 and 0.118, whereas the adapted Qwen2.5-7B gains only 0.030 and 0.078. The same pattern appears for GLM-4-9B, whose gain falls from 0.067 and 0.048 before adaptation to 0.077 and 0.020 after it. This is the expected signature of two partially overlapping mechanisms: once the adapted model already answers with railway knowledge in its parameters, the marginal value of retrieved evidence is smaller, even though the evidence still adds significantly.

The absolute ranking is what matters for the practical question. The adapted Qwen2.5-7B is the strongest generator in every language and retrieval condition, reaching 0.660 in Chinese and 0.651 in English under fused evidence, and 0.674 in Chinese under BM25. It exceeds the unadapted 14B reference by a wide margin, 0.660 against 0.442 in Chinese and 0.651 against 0.454 in English, which shows that domain adaptation is not substitutable by scale. The reverse also holds: the original Qwen2.5-7B under fused evidence reaches 0.402 and 0.440, still well below its adapted counterpart without any retrieval, so retrieved knowledge is not substitutable by adaptation either. Within a fixed hardware budget, both methods are needed to reach the best result.

A final comparison concerns the retrieval strategy itself. BM25 and fused retrieval do not order uniformly by answer quality: BM25 is ahead for Qwen2.5-7B QLoRA in Chinese (0.674 against 0.660) and for the unadapted Qwen3-14B in both languages, whereas fused retrieval is ahead for Qwen2.5-7B QLoRA in English (0.651 against 0.644) and for the original Qwen2.5-7B in both languages. The fused index has higher evidence recall, but this does not always convert into higher answer overlap, which means the final choice of retrieval strategy should be validated on the generator that will be deployed rather than on retrieval metrics alone.

4.6 Resource cost

Table VI reports the measured cost of each condition over 270 measurements per condition.

Table VI: Measured resource use over 270 generations per condition.

Model	Condition	Peak reserved GPU memory (GiB)	Mean latency (s)	Throughput (tokens/s)	Failures
Qwen2.5-7B	Original	5.49	4.01	31.9	0
Qwen2.5-7B	Adapted	16.11	6.83	17.5	0
GLM-4-9B	Original	6.76	5.65	22.7	0
GLM-4-9B	Adapted	19.49	2.63	11.9	0
Qwen3-14B (Ollama)	Unadapted	14.26	1.63	78.5	0

All five conditions completed without execution failure or out-of-memory termination, so Equation (2) holds for every configuration, including the adapted 9B model at 19.49 GiB reserved. That is the central feasibility result: the strongest quality configuration in Table V remains executable on the target workstation.

Two cautions are required when reading the memory and latency columns. Reserved memory includes caching-allocator reservations and should not be read as the memory occupied by live tensors, so the difference between the original and adapted memory figures overstates the intrinsic adapter overhead; the trainable adapters contain only 161.5M and 190.4M parameters (Table II), a small fraction of the several gigabytes

that separate the two rows. The Qwen3-14B row is measured through a different runtime and reports GPU-process memory rather than PyTorch reserved memory, so cross-backend comparison of that row is descriptive only. The GLM-4-9B adapted latency of 2.63 s is not an efficiency advantage: it reflects abnormally short or empty outputs, and the low throughput of 11.9 tokens/s is consistent with that reading, so the condition is excluded from any efficiency claim.

Figure 4 places quality and cost on the same axes. The adapted Qwen2.5-7B occupies the highest-quality position in both panels while remaining below the 24 GB boundary, which is the quality-resource trade-off this study reports. The original models are markedly cheaper, at 5.49 and 6.76 GiB, so vertical adaptation buys its quality gain at a measurable memory and latency cost rather than at no cost, and the figure is descriptive rather than a controlled estimate of adapter overhead.

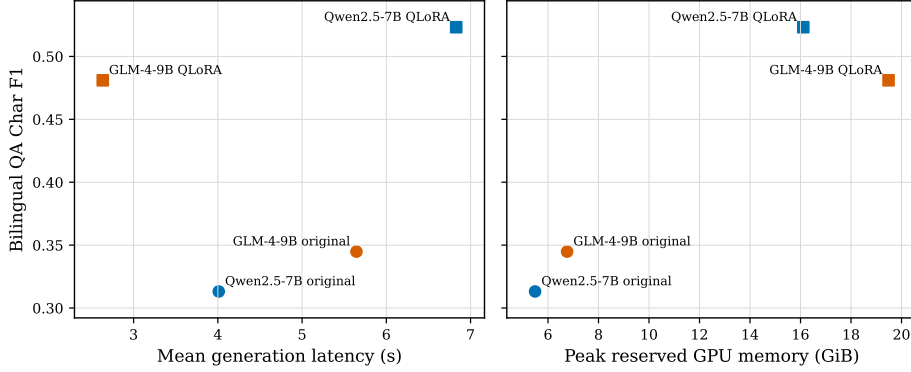


Figure 4: Quality against resource cost for the original and adapted models. Left panel: mean bilingual held-out F1 against mean generation latency. Right panel: mean bilingual held-out F1 against peak reserved GPU memory. The adapted Qwen2.5-7B holds the highest quality position in both panels while staying inside the 24 GB boundary; the original models use substantially less memory and time. Quality and resource figures are measured on separate workloads, and the Qwen3-14B reference is excluded because its runtime reports a different memory statistic. Source: Authors’ own work.

5 Discussion

5.1 Why evidence fusion helps, and where it stops helping

The retrieval results show that the two retrievers solve different parts of the bilingual access problem. BM25 is strongest for English, where queries frequently reuse the exact terminology of the evidence, while dense retrieval is strongest for Chinese, where paraphrased questions do not share surface terms with the source passage. Fusing them at rank level recovers candidates from both and produces the best evidence recall and MRR in both languages, which is the behaviour the method was chosen for: reciprocal rank fusion needs no score calibration between a lexical model and a vector model, so it can be applied to whatever pair of retrievers a domain happens to offer.

The bilingual index result is the more specific finding for vocational education. A language-specific index wins on its own language and loses badly on the other, while indexing both language fields keeps Chinese and English within a comparable range. For an institution that serves learners who ask in either language about the same knowledge item, that balance is worth more than a peak score in one language.

The limits of the method are equally clear. Fusion costs roughly three times the latency of BM25, and increasing the evidence cut-off eventually lengthens the prompt, which shifts the cost from retrieval to generation rather than removing it. The answer-quality results add a further caution: higher evidence recall did not translate into higher answer overlap in every condition, since BM25 outperformed fused retrieval for two generators. Retrieval metrics are therefore necessary but not sufficient for choosing a strategy, and the choice should be validated against the generator that will actually be deployed.

5.2 Why vertical adaptation helps, and what it does not replace

Completion-only QLoRA raises held-out railway question answering substantially in both languages for both model families. Two features of the result matter for practice. First, the gains concentrate where the original model was weakest: the Chinese scores roughly double for both models, while the English score of the already stronger GLM-4-9B barely moves. This is what would be expected if the method supplies missing

domain vocabulary and answer structure rather than uniformly improving generation. Second, the adapted 7B model outperforms an unadapted 14B model by a wide margin in both languages, which is the practical argument for vertical adaptation under a hardware constraint: at this scale, domain-specific adaptation buys more than a larger general model.

The joint evaluation is the part that distinguishes this study from single-mechanism reports. The results show that the two methods are complementary but partially overlapping. Evidence fusion improves every generator, yet the improvement shrinks as adaptation grows, from 0.141 for the original Qwen2.5-7B to 0.030 for the adapted one in Chinese. The overlap is expected, since both mechanisms supply railway knowledge by different routes, one at query time and one in the parameters, and it does not make either method redundant. Comparing the two mechanisms on equal footing for the same base model shows that vertical adaptation contributes the larger share: Qwen2.5-7B gains 0.368 in Chinese and 0.268 in English from adaptation alone, against 0.141 and 0.133 from fused evidence alone, and the same ordering holds for GLM-4-9B at 0.168 and 0.062 against 0.067 and 0.048. Adaptation and retrieval are nevertheless not interchangeable, because the best result in every language condition is obtained when the adapted model also receives fused evidence. The practical conclusion is that a vertical-domain deployment should invest in both, while expecting the marginal return of evidence fusion to fall as adaptation improves.

The same evaluation also exposes a genuine cost. The adapted models use roughly three times the reserved GPU memory of their originals and generate at roughly half the throughput. Adaptation is affordable inside a single-workstation boundary but is not free, and any deployment decision should weigh the measured quality gain in Table IV against the measured cost in Table VI.

5.3 Limitations

Three limitations bound these conclusions. The study uses one frozen corpus split, one epoch and one principal random seed, so the paired significance tests describe sample-level differences between conditions and do not estimate variation across repeated training runs. The retrieval evaluation depends on a predefined evidence-equivalence rule, so reported recall reflects that rule rather than an exhaustive relevance judgement. And the resource measurements come from a single GPU model, which means the absolute memory and latency values should not be transferred to other hardware without re-measurement, even though the qualitative ordering is likely to hold. The comparison between the PyTorch conditions and the Ollama-served reference model is descriptive for the same reason.

6 Conclusion

This study developed a railway-domain language model for bilingual vocational education under a single-workstation resource constraint, and it addressed the problem with two methods. The first is hybrid evidence fusion, which combines BM25 lexical ranking and BGE-M3 dense ranking through reciprocal rank fusion and supplies the top-ranked railway evidence as the answering context; it achieved the best evidence recall and MRR in both languages, at 0.715 Evidence Recall@5 in Chinese and 0.708 in English, and a bilingual index composition gave the best cross-language balance. The second is completion-only QLoRA vertical adaptation, which specialises a frozen 4-bit base model on reviewed railway question-answer pairs; it raised held-out character-level F1 from 0.189 to 0.398 in Chinese and from 0.437 to 0.648 in English for Qwen2.5-7B, with all paired gains significant after Holm correction. The joint evaluation showed that the two methods are complementary but partially overlapping: fused evidence improved answer F1 for all five generators in both languages, yet its marginal gain fell from 0.141 to 0.030 in Chinese once the model had been adapted, while the adapted 7B model outperformed an unadapted 14B reference by a wide margin. Every configuration ran inside the 24 GB boundary, although adaptation roughly tripled reserved memory and halved throughput. For institutions that cannot maintain multi-GPU training, these results indicate that lexical-dense evidence fusion and parameter-efficient vertical adaptation should be adopted together, and that the quality gain of each should be reported alongside its measured resource cost.

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